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Kalladeen *et al.*, *Access Microbiology* 2024;6:000746.v3
DOI 10.1099/acmi.0.000746.v3

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Group A streptococcus isolated in Guyana with reduced susceptibility to β-lactam antibiotics

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Abstract

Introduction. *Streptococcus pyogenes* [group A streptococci (GAS)] is the causative agent of pharyngitis and various other syndromes involving cellulitis, streptococcal toxic shock syndrome (STSS), and necrotising fasciitis. Although the prevalence of GAS infections globally remains high, necessitating the widespread use of β-lactam antibiotics, GAS have remained largely susceptible to these agents. However, there have been several reports of GAS with reduced susceptibility harbouring mutations in genes for penicillin-binding proteins (PBPs). The objectives of this study were to examine the *in vitro* β-lactam susceptibility patterns of group A streptococci, determine the prevalence of drug resistance, and ascertain whether such resistance could be attributed to mutations in specific PBP genes.

Methods. In this study, we sought to use Sanger sequencing to identify mutations in PBP genes of *Streptococcus pyogenes* isolated from patients that required inpatient and outpatient care that could confer reduced PBP affinity for penicillin and/or cephalosporin antibiotics. All isolates were screened for susceptibility to penicillin, amoxicillin, and cefazolin using E-test strips.

Results. While there were no documented cases of reduced susceptibility to penicillin or amoxicillin, 13 isolates had reduced susceptibility to cefazolin. Examination of *pbp1a* by Sanger sequencing revealed several isolates with single amino acid substitutions, which could potentially reduce the affinity of PBP 1A for cefazolin and possibly other first-generation cephalosporins.

Conclusion. Penicillin and penicillin-derived antibiotics remain effective treatment options for GAS infections, but active surveillance is needed to monitor for changes to susceptibility patterns against these and other antibiotics and understand the genetic mechanisms contributing to them.

DATA SUMMARY

The authors confirm all supporting data, code and protocols have been provided within the article or through supplementary data files. The genomic sequence data that support the findings of this study are openly available in GenBank (OR876961.1, OR876962.1, OR876963.1, OR876964.1, OR876965.1, OR876966.1, OR876967.1).

INTRODUCTION

Streptococcus pyogenes [group A streptococcus (GAS)] is the causative agent of numerous diseases, such as streptococcal toxic shock syndrome (STSS) and necrotising fasciitis. Some of these cases may involve multiple syndromes [1]. Invasive group A streptococcal infections are estimated to be responsible for more than 500 000 deaths annually [2, 3].

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Received 04 December 2023; Accepted 30 April 2024; Published 20 June 2024
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Keywords: antibiotic resistance; group A streptococci; Guyana; minimum inhibitory concentration; penicillin-binding proteins; *Streptococcus pyogenes*.
Abbreviations: ATCC, American Type Culture Collection; CLSI, Clinical and Laboratory Standards Institute; ECV, epidemiological cut-off value; GAS, group A *Streptococcus*; KEGG, Kyoto Encyclopedia of Genes and Genomes; MIC, minimum inhibitory concentration; PBP, penicillin-binding protein.
The genomic sequence data that support the findings of this study are openly available in GenBank (OR876961.1, OR876962.1, OR876963.1, OR876964.1, OR876965.1, OR876966.1, OR876967.1).
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While beta-lactam antibiotics have remained effective therapeutic options for *S. pyogenes*, there have been reports of reduced susceptibility to these antibiotics [3]. There have been several explanations offered for this phenomenon, which include: (a) the poor penetration of penicillin in some tissue types (e.g. tonsillar epithelial cells), which leads to intracellular *S. pyogenes* persistence [4, 5]; (b) the protection offered by β-lactamase-producing bacteria that are commonly present in the same infection sites as *S. pyogenes* (e.g. *Staphylococcus aureus*, *Moraxella catarrhalis*, *Haemophilus* spp.) [4, 6, 7]; (c) the fact that there can be coaggregation between *M. catarrhalis* and *S. pyogenes*, which improves *S. pyogenes* adherence and colonization to human epithelial cells [6]; and (d) the presence of mutations in penicillin-binding proteins (PBPs) that under selective pressure can lead to the reduced susceptibility to β-lactams [8, 9]. Although the number of reports of *S. pyogenes* with reduced susceptibility to β-lactams remains lower than other streptococci (e.g. *Streptococcus pneumoniae* and viridans group streptococci), tracking the emergence of such strains requires continuous surveillance.

To the best of the authors' knowledge, there have been no confirmed reports of *S. pyogenes* beta-lactam resistance in Guyana. However, the antimicrobial surveillance system in the country is severely lacking and there are few mechanisms in place to determine the epidemiology of invasive *S. pyogenes* infection or to effectively document the susceptibility profiles of clinical isolates countrywide, and there is no system in place for investigating the genetic mechanisms responsible for reduced drug susceptibility. Hence, this study sought to determine the β-lactam susceptibility patterns of *S. pyogenes* isolated from patients receiving inpatient and outpatient care in Georgetown, Guyana, and to determine whether any observable resistance or reduced susceptibility might be attributed to mutations in genes encoding penicillin-binding protein.

METHODS

Study location, time frame, and method of sampling

The *S. pygoenes* isolates used in this study were obtained from several sites: (a) a 600-bed tertiary care public hospital, identified as site 'A'; (b) two private general medical and surgical hospitals, identified as sites 'B' and 'C'; (c) from a fourth site, identified as site 'D', which is a multi-location private laboratory that provides testing and ambulatory care for a wide range of medical conditions. Although each of these institutions is in Georgetown, several of their patients come from the different geographical regions of Guyana.

Isolates from GAS infections were recovered at the various institutions from June 2021 to July 2022. In most instances the laboratories at each institution only employed a single method for identifying *S. pyogenes* (e.g. bacitracin susceptibility or PYR test). For this reason, all beta-haemolytic streptococci that were presumptively identified as *S. pyogenes* were included in this study.

Culture, identification, and antibiotic susceptibility testing of GAS

GAS received from the various institutions were subjected to additional testing based on the following protocol (Fig. 1): 'Conventional methods for isolation and Identification of Streptococcus pyogenes' (http://dx.doi.org/10.17504/protocols.io.rm7vzx6e2gx1/v1).

Antibiotic susceptibility testing of laboratory-confirmed *S. pyogenes* isolates was conducted using the E-test method with penicillin, cefazolin, and amoxicillin minimum inhibitory concentration (MIC) strips. Briefly, colonies of *S. pyogenes* grown overnight were used to prepare a suspension in Mueller–Hinton broth equivalent to a 0.5 McFarland standard (~1 x 10^8 c.f.u. ml^{-1}). A sterile swab was dipped into the broth and used to streak the surface of a culture plate (90×15 mm) containing Mueller–Hinton (MH) agar with 5% sheep blood. Using aseptic techniques, the MIC strips were gently pressed onto the surface of the agar. The plates were incubated in an inverted position at 35–37°C in 5% CO_2 atmosphere for 24 h. Quality control for this phase of testing was performed using *S. pyogenes* ATCC 19615. The minimum inhibitory concentration was identified as the point where the edge of the inhibition ellipse intersected the strip. The MIC results were interpreted per the CLSI M100-S31 guidelines [10].

DNA extraction and Sanger sequencing

S. pyogenes isolates with reduced β-lactam susceptibility were processed using a crude DNA extraction protocol to prepare them for sequencing. Briefly, five to seven colonies of overnight growth of *S. pyogenes* were suspended in 1 ml of molecular-grade water in a DNase/RNase-free micro-centrifuge tube. The micro-centrifuge tube was then boiled at 100°C for 10 min in a water bath before being centrifuged for 5 min at 1 000 r.p.m. The supernatant was removed and placed into a DNase/RNase-free cryogenic vial and stored at −20°C. The frozen sample was then shipped to GENEWIZ (RRID:SCR_003177) for further analysis using Sanger sequencing. Using their proprietary methodology, they performed polymerase chain reaction (PCR) on purified genomic DNA, amplifying an internal region <1 kb. Using a template, they sequenced a target region of the PCR transcript. For the identification of substitutions and deletions in the amino acid sequence of *pbp1a* and *pbp2x*, NCTC12064 (GenBank accession no. NZ_LS483338.1) was used as the *S. pyogenes* reference strain.

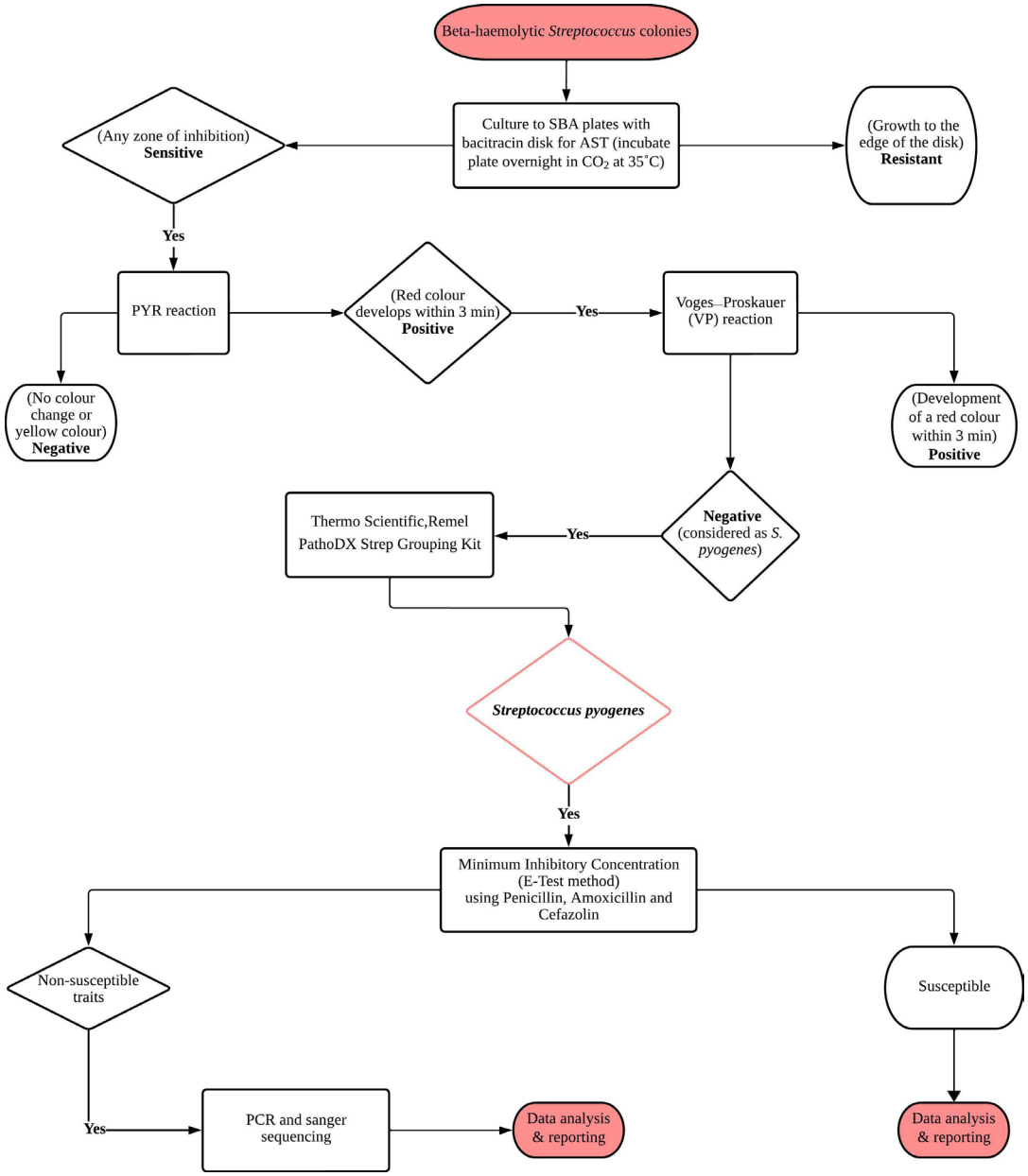


Fig. 1. Algorithm for identifying *S. pyogenes*.

Data analysis

The MIC_{50} (i.e., MIC at which 50% of isolates were inhibited) and MIC_{90} (i.e. MIC at which 90% of isolates were inhibited) were calculated based on the susceptibility test results. The CLSI guidelines do not provide clinical breakpoints for cefazolin and, as such, an assessment of the MIC distribution for this agent was performed using an epidemiological cut-off value (ECV) as described by other authors [11]. Briefly, the 24 h MIC distributions for cefazolin from this study were collected and used to determine the modal MIC. The MIC distributions were then aggregated and used to establish the wild-type MIC distributions and ECVs. To the best of our knowledge, this was the first study to investigate infections caused by GAS and, currently, in Guyana, there is no national database for collecting MIC data for GAS. Therefore, we assumed that the ECV would be the MIC that encompassed ≥95% of all MIC values in the wild-type MIC distributions [11].

Sequencing results were analysed using NCBI BLAST and sequence alignment for the detection of substitutions and deletions was performed using SnapGene v. 7.0.2 (RRID:SCR_015052). The Kyoto Encyclopedia of Genes and Genomes (KEGG) GENES database (https://www.genome.jp/kegg/genes.html) was used to define the transglycosylase and transpeptidase domains of the

PBPs. SPy_1649 was used as a reference for PBP 1A. Sequences with amino acid substitutions or deletions along the PBP gene were considered to be possible candidates for lower β-lactam affinity and contributing to increased MICs for these antibiotics.

RESULTS

Bacterial isolates

Fifty-three beta-haemolytic isolates were presumptively identified as *S. pyogenes* from three of the four institutions that participated in this study. Most of these isolates were obtained from site A (n=42). The other institutions contributed 11 isolates (9 from site D, 2 from site B, and none from site C). Of the 53 isolates, only 37.7% (n=20) were confirmed as *S. pyogenes* using conventional identification methods.

Antimicrobial susceptibility testing

All 20 laboratory-confirmed *S. pyogenes* isolates were subjected to antibiotic susceptibility testing using the E-test method with penicillin, cefazolin, and amoxicillin. All isolates were susceptible to penicillin and amoxicillin (Table 1).

The MIC for penicillin ranged from 0.016 to 0.032 µg ml^{-1} for all the strains (n=20). The MIC_{50} and MIC_{90} were 0.016 and 0.023 µg ml^{-1}, respectively. For amoxicillin, the MICs observed ranged from 0.016 to 0.064 µg ml $^{-1}$. The MIC_{50} and MIC_{90} values were both 0.023 µg ml.

For cefazolin, there were seven isolates that had MICs of 0.12 µg ml^{-1}. Thirteen isolates had higher MICs, ranging from 0.19 to 0.75 µg ml^{-1}. As there was no clinical breakpoint for cefazolin for *S. pyogenes*, we investigated the wild-type MIC distribution of cefazolin using the MIC of the reference organism, i.e. *S. pyogenes* ATCC 19615 (MIC=0.09), and all the isolates tested. This was done because of the absence of published or reported MICs for group A streptococci locally. Using a 95% epidemiological cut-off value (ECV) [11], the wild-type ECV was determined to be 0.19. All 13 isolates had MICs ≥0.19 and were considered to have

Table 1. Source and β-lactam susceptibility profile for laboratory-confirmed GAS

Clinical isolate	Source facility	Specimen type	Antibiotic agents (MIC, mg ml^{-1})		
			Penicillin S≤0.12	Cefazolin	Amoxicillin S≤0.12
S-001	A	Pharyngeal swab	0.016	0.125	0.023
S-002	A	Urine	0.023	0.19	0.064
S-003	A	Wound	0.023	0.125	0.016
S-004	A	Urine	0.016	0.19	0.016
S-006	A	Pharyngeal swab	0.016	0.38	0.016
S-010	A	Pharyngeal swab	0.016	0.19	0.016
S-017	A	Pharyngeal swab	0.016	0.19	0.023
S-023	A	Pharyngeal swab	0.023	0.19	0.016
S-026	A	Pharyngeal swab	0.016	0.125	0.023
S-027	A	Wound	0.016	0.19	0.023
S-030	A	Pharyngeal swab	0.016	0.19	0.023
S-035	D	Blood	0.016	0.25	0.023
S-036	D	Wound	0.023	0.75	0.016
S-037	D	Urine	0.023	0.125	0.023
S-046	A	Pharyngeal swab	0.016	0.125	0.023
S-047	A	Throat swab	0.016	0.25	0.016
S-048	A	Sputum	0.032	0.125	0.016
S-049	D	Sputum	0.016	0.19	0.023
S-050	D	Sputum	0.023	0.25	0.023
S-051	D	Sputum	0.016	0.125	0.023

Table 2. Deduced amino acid substitutions in the PBP1a gene of the 13 isolates identified as *Streptococcus pyogenes*

Clinical isolate	Position of mutation in *pbp1a*
S-002	–
S-004	–
S-006	H291N, I337V, I361T
S-010	–
S-017	–
S-023	–
S-027	I361T
S-030	T556A
S-035	H291N, I337V, I361T
S-036	H291N, I337V, I361T
S-047	–
S-049	H291N, I337V, I361T
S-050	I486M, I645M

reduced susceptibility to cefazolin when compared to the wild-type strains. Eight of these isolates were recovered from patients hospitalised at site A, while five were from site D.

PCR and Sanger sequencing

Based on the results from susceptibility testing, the 13 isolates adjudged to have reduced cefazolin susceptibility were further assessed to understand whether the reduced drug susceptibility was linked to genetic modifications in their penicillin-binding proteins. These isolates were subjected to crude DNA extraction, and the extracted DNA was sent to GENEWIZ (RRID:SCR_003177) for DNA purification, PCR, and Sanger sequencing of the PBP 1A and PBP 2X genes.

The results revealed that there were substitutions in the amino acid sequence of the transglycosylase-transpeptidase region of *pbp1a* (Table 2). These substitutions were considered as possible causes of the reduced PBP affinity for β-lactams.

Unfortunately, sequencing of *pbp2x* could not be performed because degradation of the primary genomic sequences prevented reliable PCR and Sanger sequencing.

DISCUSSION

This study describes, to the best of our knowledge, the first reported cases of group A streptococci that exhibited reduced susceptibility to a β-lactam antibiotic. Based on the results obtained, it is evident that penicillin and amoxicillin were the most effective treatment options for *S. pyogenes* infections. These results were not entirely unexpected, as they were similar to those of Camara *et al.* [12] and several other studies performed in other parts of the world, including Morocco [13], France [14], and Germany [15]. However, of the 20 laboratory-confirmed *S. pyogenes* isolates identified, 13 were deemed to express reduced susceptibility to cefazolin when compared to the wild-type strain. This finding was somewhat unusual, given that cephalosporins have been described as being superior to penicillin antibiotics for GAS infections [16]. Cefazolin is recommended for treating respiratory infections with *S. pyogenes* in adults and paediatric patients [17]. First-generation cephalosporins, like cefazolin and cephalexin, also represent effective first-line therapeutic alternatives to penicillin for GAS infections in patients at risk of anaphylaxis due to penicillin [18, 19]. Although the CLSI interpretive guideline does not provide a clinical breakpoint for cefazolin and, given that this was the first research to examine cefazolin susceptibility among GAS in Guyana, an epidemiological cut-off that encompasses 95% of the wild-type strains was considered. Therefore, we assumed that all isolates that had MICs below the ECV were 'susceptible' and those with MICs at or above the cut-off were assumed to have reduced susceptibility relative to their wild-type counterparts. This approach was based on work done by several other authors [11, 20, 21].

Among the proposed mechanisms contributing to reduced susceptibility of streptococci to cephalosporins and other β-lactam antibiotics, the role of penicillin-binding proteins (PBPs) has received much attention. In particular, the alteration of PBPs resulting in their low affinity for these antibiotics or the acquisition of new PBPs have been identified as key resistance determinants in *S. pneumoniae* and viridans streptococci [22–25]. While research has shown that cephalosporin resistance is linked to alterations

to PBPs 1A and 2X [26], mutations of these PBPs in *S. pyogenes* are not widespread [27], although there have been reports that identified a single amino acid mutation of *pbp2x* as a mediator of increased MICs for *β*-lactam in *S. pyogenes* [28, 29]. Furthermore, it has been shown that alterations in PBP1A can help to potentiate low- and high-level *β*-lactam resistance in other streptococcal species [30, 31].

To investigate the role of mutations in *pbp1a* and *pbp2x* in mediating reduced susceptibility to cefazolin, the 13 isolates were subjected to PCR and Sanger sequencing. We found that seven of the isolates (GenBank accession nos OR876961.1, OR876962.1, OR876963.1, OR876964.1, OR876965.1, OR876966.1, OR876967.1) had amino acid substitutions near to and within the conserved active motif of *pbp1a*, with four of these isolates having the same amino acid substitutions (i.e. histidine for asparagine at position 291, isoleucine for valine at position 337, and isoleucine for threonine at position 361). It is conceivable that these amino acid substitutions caused the reduced affinity of the PBP for cefazolin but not penicillin. This is not unusual, given that point mutations that affect the affinity of a PBP for one *β*-lactam antibiotic may have little or no effect for another [28]. Furthermore, reduced susceptibility to cephalosporins is more commonly associated with alterations in both *pbp1a* and *pbp2x* [26]. Unfortunately, we were unable to sequence *pbp2x*.

Interestingly, the similarities between the amino acid substitutions in the four isolates suggests that these isolates could be closely related, thus representing a prevalent clone. However, further investigation is required to better understand and explain these findings. It seems entirely possible that repeated and poorly regulated antibiotic use may be essential factors driving reduced cephalosporin susceptibility. These factors are likely to select for GAS that have progressively accumulated amino acids substitutions in PBPs [9, 22].

This is the first study, as far as the available literature suggests, to identify mutations in the *pbp1a* in GAS isolated from patients receiving inpatient and outpatient care in Guyana that likely caused reduced susceptibility to a first-generation *β*-lactam cephalosporin antibiotic in the country. These findings reinforce the importance of conducting active monitoring of *β*-lactam susceptibility among these isolates and utilising tools such as molecular testing to better understand the epidemiology of drug resistance and to study other genetic mechanisms contributing to it.

Funding information
This work received no specific grant from any funding agency.

Acknowledgements
We acknowledge the laboratory teams at sites A and D and thank them for providing us with assistance and the space needed to conduct all the laboratory work.

Author contributions
M.K. and P.E.A. were responsible for conceptualisation. M.K. was responsible for method development, investigations, formal analyses of initial test data and writing of the original draft. P.C. was responsible for method validation, formal analysis of molecular data, data curation, and writing of the final draft. P.E.A. was responsible for study supervision. All authors read and approved the final draft of the study.

Ethical statement
Ethical approval for this study was granted by the Research Ethics Committee of the University of the West Indies, St Augustine Campus, Trinidad and Tobago. Additional permission was granted by the Institution Review Board of the Ministry of Health, Guyana and the hospitals and clinics where the isolates were collected.

Conflicts of interest
The authors declare that there are no conflicts of interest.

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